MTH441: Lab 5

Questions

1. Let us first generate some data from a Logistic Regression Model:

```
set.seed(123)
n <- 50
p <- 4

# Making model matrix
Xtilde <- matrix(rnorm(n*p), ncol = p, nrow = n)
X <- cbind(1, Xtilde)

# True betas
beta.truth <- 1:(p+1)

# calculating probability of success
pi <- 1/(1 + exp(-X %*% beta.truth))

# generating Bernoulli response
y <- rbinom(n, 1, prob = pi)
```

Answer the following questions:

a. Using the Newton-Raphson algorithm and coding the gradient and Hessian by yourself, write a function that calculates the MLE of β.

b. Estimate the approximate variance of the MLE of β

c. Perform individual z-tests for all 5 regression coefficients.

d. Using the following code that implements the GLM fitting, match your values in (a), (b), and (c) with the table of coefficients below:

```
## removing intercept since X already has intercept
fit <- glm(y ~ X - 1, family = "binomial")
summary(fit)
```

2. We will plot the log-likelihood of the logistic regression model to better understand the concavity of the log-likelihood. In order to draw this properly, we will need to construct a 1-covariate example (no

intercept). Let us consider an artificial dataset where we have the following observed.

```
x <- c(-2, -1, -3, -4,  1,  2,  3,  4)
y <- c( 0,  1,  0,  0,  1,  1,  1,  1)
```

The following code calculates the log-likelihood at a given value of β for a given x and y:

```
# Stable log-likelihood for logistic regression
loglik <- function(beta, x, y) {
  eta <- x * beta
  # use log1p for numerical stability: log(1 + exp(eta))
  sum(y * eta - log1p(exp(eta)))
}
```

Plot the log-likelihood of β in the window $[-5, 10]$. Where is the log-likelihood maximized?

3. Repeat the above for the following data:

```
x <- c(-2, -1, -3, -4,  1,  2,  3,  4)
y <- c( 0,  0,  0,  0,  1,  1,  1,  1)
```

What do you observed for the above data? Mathematically observe the reason for this behavior?

4. We will now repeat Q1 again with data generated from a different seed, but just use the `glm` function. What do you observe? Based on your answer for Q3, can you explain what is happening here?

```
set.seed(44)
n <- 50
p <- 4

# Making model matrix
Xtilde <- matrix(rnorm(n*p), ncol = p, nrow = n)
X <- cbind(1, Xtilde)

# True betas
beta.truth <- 1:(p+1)

# calculating probability of success
pi <- 1/(1 + exp(-X %*% beta.truth))

# generating Bernoulli response
y <- rbinom(n, 1, prob = pi)
fit <- glm(y ~ X - 1, family = "binomial")
summary(fit)
```

5. On April 10, 1912, the RMS Titanic sank after colliding with an iceberg on its maiden voyage. The accident killed 1502 of the 2224 passengers and crew on board. A subset of the data of the survival of passengers and crew can be loaded in `R` using

```r
titanic <- read.csv("https://dvats.github.io/assets/data/titanic.csv")
```

The dataset has the response: whether a passenger survived the tragedy (`Survived`). Additional information includes the sex of the passenger (1 if male, 0 otherwise)(`Sexmale`), age (`Age`), the number of siblings/spouse aboard (`SibSp`), the number of parents/children aboard (`Parch`) and the passenger's fare (`Fare`) (in British pounds). The data-set contains 712 observations.

a. Fit a logistic regression model for this dataset.

b. Which is the coefficient that most significantly impacts the probability of survival?

c. Provide an interpretation for the regression coefficient of `Fare`.

d. In James Cameron's 1997 directorial venture, "Titanic' ', it is often debated that the character of Jack Dawson (played boisterously by Leonardio DiCaprio) could have probably survived if Rose Bukater (played by the ever-brilliant Kate Winslet) had made room for him on the raft. Let's end this debate once and for all.

   Jack Dawson was 20 years old when the tragedy happened. He boarded the ship with no family or spouse and paid 7.5 British pounds for his ticket. Rose Bukater was 19 years old on the tragic day. She boarded the ship with her fianceé (treat this as a spouse) and her mother. She paid 512 British pounds for her ticket. What are the estimated probabilities of survival for Jack Dawson and Rose Bukater?